Poisson Deconvolution

Reconstructing Fatal Infection Curves

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November 2026

1 Abstract

This team study reconstructs the timing of infections that eventually produced observed English hospital COVID-19 deaths. A positive spline curve is passed through an assumed infection-to-death delay distribution and fitted with a penalised Poisson likelihood. Fakhruddin Hakim Hussain contributed bootstrap inference, plotting and the write-up; Mansi Neb contributed matrices and likelihood/gradient work, and Riccardo Monforte contributed fitting and smoothing selection. The source records approximately equal overall effort.

The recovered input contains 150 daily counts, totalling 29,442 hospital deaths. A new independent Python implementation checks the archived R equations, rather than claiming an execution of that R script. It evaluates all fifty smoothing candidates and performs 200 fixed-smoothing weighted-case bootstrap refits. The reconstructed fatal-infection peak precedes the fitted death peak by twenty days.

All fifty candidate fits and all 200 bootstrap fits meet the stated gradient threshold. The initial analytical-gradient check has maximum central-difference error 1.39×10^{-6}. These checks support the implementation, not the validity of the assumed delay distribution or nominal interval coverage.

The latent curve concerns fatal infections linked to these records, not all community infections. The research decision is to retain the reconstruction as a conditional inverse-problem study and assess delay, dependence and smoothing uncertainty before any stronger epidemiological interpretation.

2 Observation Process and the Target

Deaths occur after infection, with varying delays. Simply shifting every observed death by one fixed number of days cannot represent that spreading process. Let y_i be the count on day t_i and $f(t)$ the expected daily infections that lead to the recorded deaths. The model is

$$y_i \sim \text{Poisson}(\mu_i), \qquad \mu_i = \sum_{j=1}^{\min(29+i,80)} \pi(j) f(t_i - j). \tag{1}$$

The source assumes $\log d \sim \mathcal{N}(3.151, 0.469^2)$, evaluates its density at delays one to eighty days and renormalises the weights. Early summation limits truncate the available prehistory. This is a modelling convention, not independent estimation of the delay law from the deaths.

The input covers days 62–211 of leap year 2020. The infection grid extends thirty days before the first observed death day, giving 180 grid points. Because the delay kernel sums to one and no infection-fatality ratio is estimated, f does not identify the total infection burden. A misleading source comment attributing peak height to a fatality fraction has been corrected.

3 Spline Model, Likelihood and Gradient

Eighty cubic B-splines form the infection basis $\widetilde{X}$. Convolving each basis column with the delay weights gives the death design matrix X. Positive coefficients use $\beta = \exp\gamma$ and $f = \widetilde{X}\beta$. A second-difference matrix D defines $S = D^\top D$.

For nonnegative row weights w_i, the implemented objective, omitting Poisson factorial constants, is

$$Q(\gamma) = \sum_i w_i\{\mu_i - y_i \log \mu_i\} + \frac{\lambda}{2}\beta^\top S\beta, \qquad \mu = X\beta. \tag{2}$$

Its gradient is

$$\nabla_\gamma Q = \beta \odot \left[X^\top\{w \odot (1 - y/\mu)\} + \lambda S\beta\right]. \tag{3}$$

Positivity is enforced through the coefficients, rather than by clipping the reconstructed curve after fitting. Numerical floors protect the log likelihood. Gradient agreement is a numerical implementation check and does not establish a unique global optimum.

4 Smoothing Selection and Numerical Checks

The archived grid contains fifty values of $\log \lambda$ from -13 to -7. The criterion is twice the unpenalised negative log likelihood plus $\log(n)$ times approximate effective degrees of freedom. The latter uses

$$\text{edf} = \text{tr}\{(H_0 + \lambda S)^{-1} H_0\}, \quad H_0 = X^\top \text{diag}(y_i/\mu_i^2) X. \tag{4}$$

This is the source's local observed-curvature approximation, not an exact dimension or a held-out predictive score. The negative criterion values omit constants and are meaningful only for comparing these fits on the same data.

The independent review uses SciPy BFGS, a maximum of 2,000 iterations and a stationarity threshold of 10^{-3} on the maximum absolute gradient. Every grid fit meets that threshold. The selected value is $\lambda = 1.85615 \times 10^{-4}$, with approximate edf 19.22 and maximum gradient 5.18×10^{-6}. It lies within the searched grid. Different optimisers and tolerances mean this is not a bit-for-bit reproduction of the original R optimisation.

5 Reconstruction and Bootstrap Results

The reconstructed fatal-infection curve peaks at day 79; fitted deaths peak at day 99. The twenty-day gap is conditional on a preassigned delay kernel and the smoothing model, rather than a new estimate of a universal biological delay. In-sample death-count RMSE is 17.37; no predictive holdout was used.

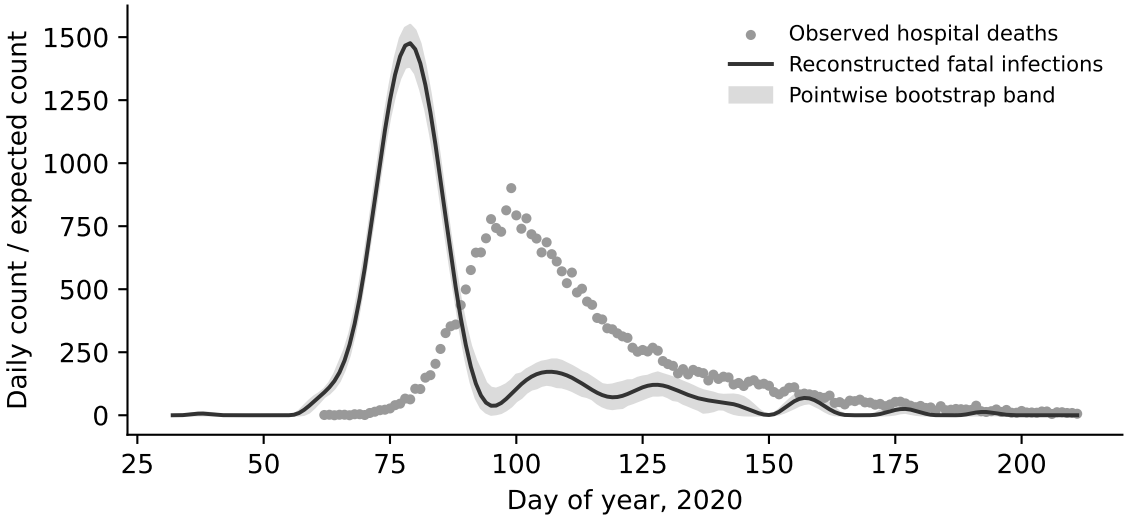

Figure 1: Independent Python reconstruction on the recovered aggregate input. The band uses 200 fixed-smoothing weighted-case bootstrap refits; it is pointwise and conditional on the assumed observation model.

Each bootstrap draws 150 row indices with replacement and uses their counts as likelihood weights, leaving the time/design rows in place. It refits at the selected smoothing strength. Seed 20261101 is recorded. All 200 fits have maximum gradient below 10^{-3}; the largest is 1.81×10^{-5}. Reported percentiles use all fits. They do not include re-selection of smoothing, uncertainty about the delay law, serial dependence or simultaneous coverage over the curve.

6 Conclusion, Limits and Reproducibility

The work demonstrates likelihood construction, an analytical gradient, regularisation and bootstrap refitting for an ill-posed inverse problem. The user contribution is retained explicitly as part of the team effort. The new reference implementation supplies an auditable numerical check of that model.

A better-fitting death curve does not uniquely identify the latent infection curve. Endpoint behaviour, the thirty-day prehistory, overdispersion and a fixed delay law matter. The study does not forecast all infections, identify an infection-fatality ratio or establish a policy effect. A next study should compare plausible delay laws, study residual dependence, include smoothing selection inside uncertainty analysis and test recovery on synthetic curves with known truth.

The recovered input and its hash identify the replay. The original R script was not run in this environment. The public Python runner accepts an explicitly supplied input file and exports aggregate checks; private source rows and local reconstructed-curve files are not redistributed as repository datasets.

References

[1] Project authors. Pinned source code and evidence. Original study distinguished from the November 2026 audit.

[2] Companion verification record. Seed, inputs, numerical scope and limitations.

[3] Practical 3: Smooth deconvolution. Assignment specification supplied with the original input and script. The model construction and target follow that specification.